

Practical 1 Problem Statement Manufacturing Production Management System (MPMS)

Aim

To define a clear and comprehensive problem statement for the software project — “Manufacturing Production Management System”.

1. Introduction

In modern manufacturing industries, production management plays a critical role in ensuring timely delivery, efficient machine utilization, and smooth coordination between workers and supervisors. Many medium-scale factories still rely heavily on manual scheduling and production tracking methods.

In a manufacturing plant producing precision cutting tools, multiple machine types such as CNC cutting machines, drilling machines, and grinding machines are used. Some machines are identical while others are specialized for particular operations.

Currently, production scheduling and machine assignment are handled manually by supervisors based on experience rather than a structured system. This often causes inefficiencies, delays, and poor resource utilization.

Therefore, there is a need for an automated Production Management System that can intelligently manage orders, workers, machines, and production scheduling.

2. Problem Description

- Orders are received manually through calls, emails, or direct communication.
- Production schedules are maintained manually.
- Supervisors assign machines based on personal judgment.
- Urgent orders disrupt already running schedules.
- Machine workload balancing is not optimized.
- Real-time machine availability is difficult to track.
- Workers and supervisors lack centralized coordination.
- Multi-stage operations are difficult to manage efficiently.
- No centralized dashboard exists for monitoring production flow.

These issues result in production delays, poor machine utilization, workflow interruptions, and reduced productivity.

3. Proposed Solution

The Manufacturing Production Management System (MPMS) is a web-based software solution designed to automate production scheduling and machine allocation. The system provides role-based access for Admin, Supervisor, and Worker users.

4. Scope

- Designed for medium-scale manufacturing plants.
- Supports multiple machine types.
- Handles production sequencing operations.
- Supports urgent and priority-based order processing.
- Accessible through web browsers.
- Real-time production monitoring supported.

5. Objectives

- Automate production order management.
- Improve machine utilization efficiency.
- Reduce manual scheduling dependency.
- Improve coordination between supervisors and workers.
- Minimize production delays.

6. Expected Outcome

The system will provide automated machine allocation, better workflow coordination, reduced production delays, and real-time visibility into machine status and workload.

Hardware & Software Requirements

Category	Requirement
Operating System	Windows 10/11 or Linux
Processor	Intel i5 or higher
RAM	8 GB minimum
Database	MySQL / PostgreSQL
Backend	Node.js / Python
Frontend	React.js / HTML-CSS-JS
Web Server	Apache / Nginx
Browser	Chrome / Firefox / Edge